Real-Time Image Sharpening for Video Conferencing Using Knowledge Distillation

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Abstract

A lightweight model was trained to improve frames degraded by 2x bicubic downscaling and upscaling, as specified in the Intel Unnati problem statement. A pretrained SwinIR model supplied teacher targets to a 2,380-parameter convolutional student. Across 125 untouched test images, mean RGB SSIM increased from 0.9040 for the degraded input to 0.9175 for the student. At 1920x1080, OpenVINO GPU processing averaged 48.03 FPS. In a blinded study with ten volunteers, the output received a mean opinion score of 3.735/5 versus 3.765/5 for the input; the difference did not show a subjective preference. The implementation demonstrates real-time processing under the specified simulation, while its visible improvement and generalization to actual network artifacts remain limited.

Keywords: image sharpening, knowledge distillation, video conferencing, SwinIR, OpenVINO

1. Introduction

Low-bandwidth video conferencing can reduce transmitted resolution and visible detail. The Intel Unnati brief [1] proposes a teacher-student image-sharpening model that should process 1920x1080 frames at 30-60 FPS or higher. It specifies bicubic or bilinear downscale/upscale as a controlled way to create lower-quality training inputs, plus SSIM, a diverse test set of more than 100 images, and a subjective MOS study.

This paper reports an implementation of that specified setup. The task is narrower than repairing every poor internet call: compression blocks, frame loss, camera focus, and motion blur are not modeled. We therefore report objective quality, speed, and human ratings separately.

2. Related Work

DIV2K [2] provides high-resolution images for super-resolution experiments. SwinIR [3] is a pretrained transformer-based restoration model and serves as the teacher here. Restormer [4] is another high-resolution restoration method; it is not the model used in our training. The student uses depthwise convolution to keep computation small.

3. Method

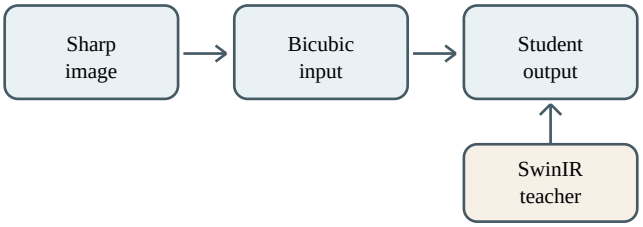


Fig. 1. Controlled degradation and teacher-guided student training. The teacher sees the low-resolution crop; the student sees the bicubic enlarged input.

A sharp 256x256 crop is reduced by 2x, then enlarged to its original size using bicubic interpolation. The original crop is the ground truth. The student predicts a correction that is added to the enlarged input. It processes features at half resolution using 16 channels and four depthwise residual blocks, then returns a full-resolution image through pixel shuffle. The resulting network has 2,380 trainable parameters.

The teacher is the official pretrained SwinIR-S x2 model [3]. Training minimizes L1 reconstruction error to the sharp crop, a weighted gradient L1 term, and a weighted L1 term to the teacher output. Adam uses an initial learning rate of 0.0002 with cosine annealing. The final student was trained for 12 epochs on a Colab Tesla T4 with batch size four and seed 2110. The best downloaded checkpoint was selected at epoch 11 using a 50-image training holdout.

4. Evaluation Protocol

Training used 750 DIV2K images. Final quality testing used 100 untouched DIV2K validation images and 25 additional images: five each from text, nature, people, animals, and games. Each test image underwent the same 2x bicubic degradation. PSNR and SSIM were computed in RGB on 8-bit images after a two-pixel border crop; the SSIM data range was 255.

For subjective testing, 20 native-resolution detail crops were selected from untouched validation images. The degraded input and student output were shown in randomized, blinded A/B order. Ten volunteers rated both sides separately from 1 (very poor) to 5 (excellent); equal ratings were allowed.

5. Results

The student exceeded the 0.90 mean SSIM target on the 125-image test (Table 1). The gain was 0.0135 SSIM and 0.6872 dB PSNR over the degraded input. These averages establish measurable improvement under the controlled degradation, although individual images varied.

Method	SSIM	PSNR (dB)
Bicubic input	0.9040	31.0119
Student output	0.9175	31.6990

Table 1. Mean full-RGB quality on 125 test images.

On the 100 DIV2K validation images alone, the student reached 0.9152 mean SSIM. Table 2 shows category-level means. The full per-image scores are provided with the project files.

Category	Input	Student
DIV2K (100)	0.9014	0.9152
Text (5)	0.9089	0.9222
Nature (5)	0.8884	0.9044
People (5)	0.8978	0.9129
Animals (5)	0.9733	0.9769
Games (5)	0.9025	0.9170

Table 2. Mean RGB SSIM by test category; counts are in parentheses.

5.1. Full-HD Speed

The final model was exported to OpenVINO for 1920x1080 BGR frames. On a Lenovo Yoga Slim 6i (Intel Core i5-13500H, 16 GB RAM, Iris Xe GPU), batch-one inference was timed after ten warmups over 50 runs. Engine latency averaged 18.09 ms; full frame processing averaged 20.82 ms, or 48.03 FPS (Table 3). Capture, display, and encoding were excluded.

Measurement	Mean	FPS
Engine	18.09 ms	55.29
Frame processing	20.82 ms	48.03

Table 3. GPU timing at 1920x1080, batch size one.

Two screenshots of the webcam demonstration showed 35.8 and 40.6 processing FPS. They are point readings, not a sustained average. The displayed preview was scaled down; the model itself processed full-HD frames.

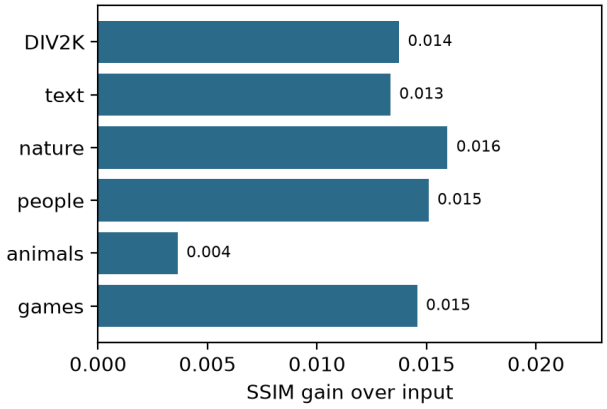


Fig. 2. Mean RGB SSIM gain over the degraded input for each test category.

5.2. Subjective Study

The ten volunteers each scored all 20 blinded pairs, yielding 200 paired ratings. Mean opinion score was 3.735/5 for student output and 3.765/5 for degraded input (Table 4). In 43 pairs the student was rated higher, 108 were tied, and 49 favored the input. The mean paired difference was -0.030 points. The study therefore does not show a perceived preference for the output.

Image	MOS	SD	Ratings
Input	3.765	0.665	200
Student	3.735	0.653	200

Table 4. Human ratings on 20 blinded image pairs; SD is sample standard deviation.

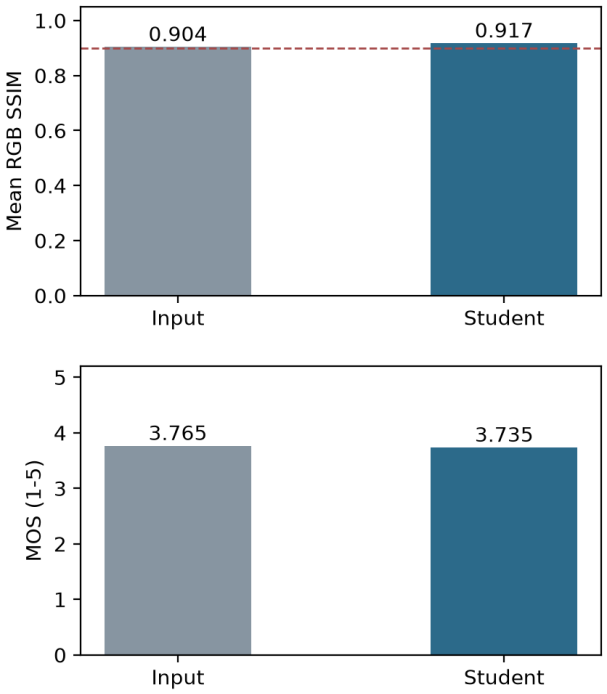


Fig. 3. Objective SSIM improves while volunteer MOS is nearly unchanged. The dashed line marks the 0.90 SSIM target.

The quantitative and subjective results answer different questions. A model can improve average pixel structure without making a clearly preferable picture at normal viewing size. The small human sample also limits conclusions about visual preference.

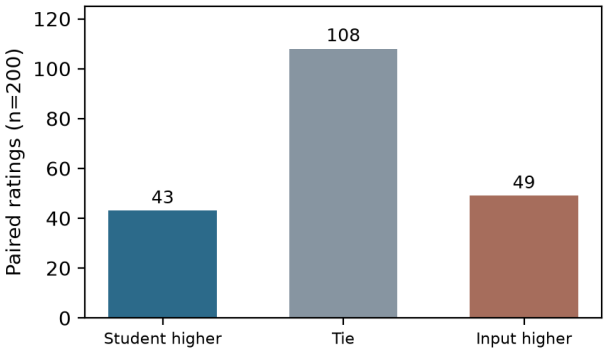


Fig. 4. Counts of pairwise volunteer ratings. More than half of the comparisons were ties.

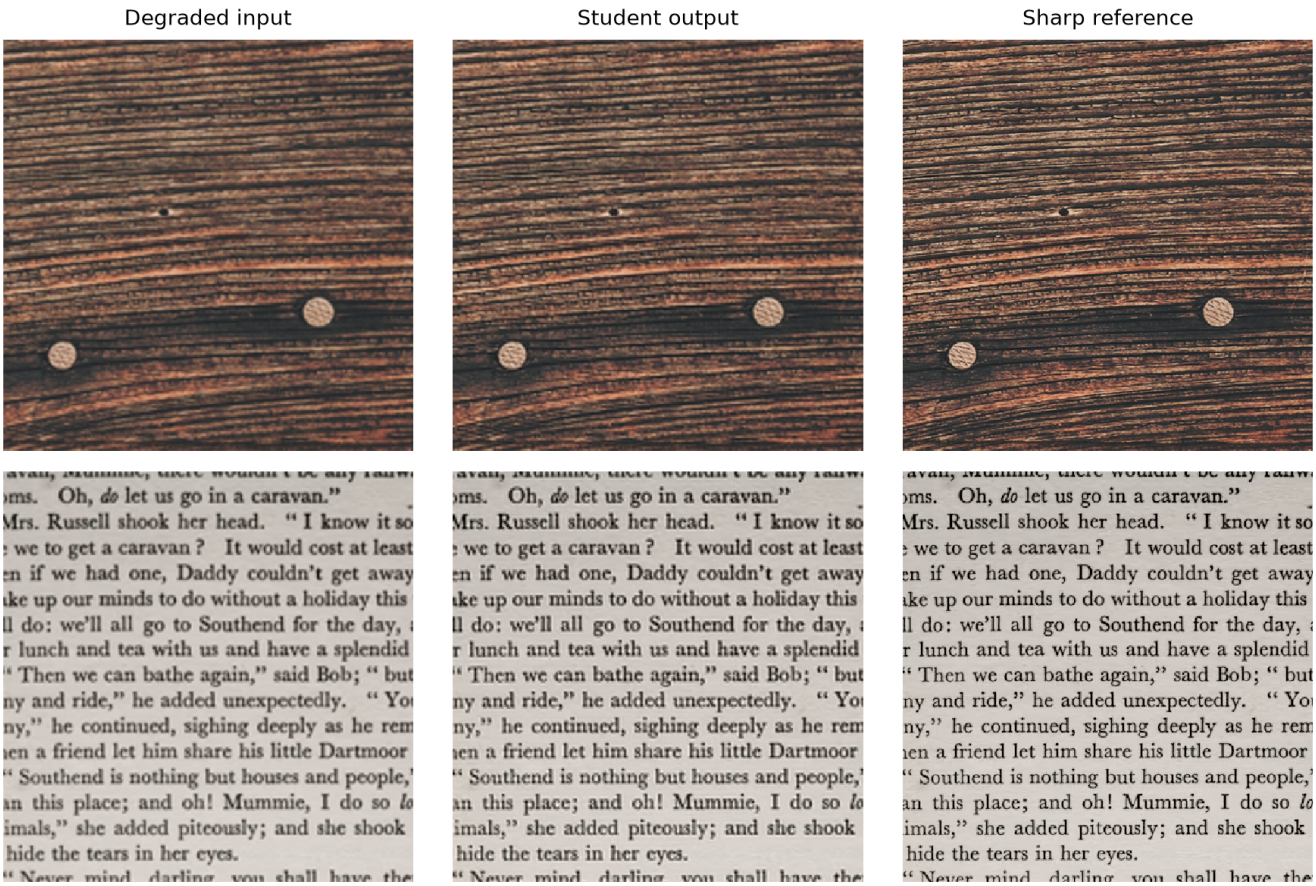


Fig. 5. Native-resolution matched crops: degraded input, student output, and sharp reference. Top: DIV2K image 0885 (wood texture); bottom: a text-category test image. These examples were selected for detail and are not a substitute for the 125-image averages in Table 1. No extra sharpening was applied to the displayed outputs.

6. Discussion and Limitations

The experiment met the specified image count, mean SSIM threshold, and measured full-HD processing speed. Qualitative differences are subtle, and the MOS study did not find a preference for the student output. The teacher-guided training procedure was implemented, but these measurements alone do not isolate how much benefit came specifically from teacher guidance.

The training simulation models only bicubic resolution loss. Real conferencing streams can also contain codec artifacts, dropped frames, low light, focus blur, and motion blur. The model processes frames independently, so temporal flicker is possible. The current app is a standalone webcam/video demonstration rather than a conferencing virtual camera.

7. Conclusion

A 2,380-parameter student can process full-HD frames in real time on the tested Intel laptop while improving mean SSIM on a controlled 125-image benchmark. Human ratings show that this gain was not clearly visible in the 20-pair study. Future work should test realistic compressed video and temporal consistency before claiming improvement in actual low-bandwidth calls.

8. Reproducibility

The submission archive contains the Python source, trained checkpoint, OpenVINO model, per-image and category-level metrics, benchmark JSON, and anonymized MOS responses with the blinded answer key. The official DIV2K images and supplementary test images are not bundled because they are large and may have separate distribution terms. Code defaults and exact run commands are documented in the accompanying README.

The 125-image test scores were measured from the downloaded best student checkpoint. The qualitative figure uses native-resolution crops and the actual student output. No display-only sharpening was applied. The plotted MOS values come from the ten recorded participant files.

References

[1] Intel Unnati Industrial Training 2025, Problem Statement 2, Image Sharpening using knowledge distillation.

[2] E. Agustsson and R. Timofte, NTIRE 2017 Challenge on Single Image Super-Resolution: Dataset and Study, CVPR Workshops, 2017.

[3] J. Liang et al., SwinIR: Image Restoration Using Swin Transformer, ICCV Workshops, 2021.

[4] S. W. Zamir et al., Restormer: Efficient Transformer for High-Resolution Image Restoration, CVPR, 2022.